

Tumor's other immune targets: dendritic cells

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Abstract: The induction of apoptosis in T cells is one of several mechanisms by which tumors escape immune recognition. We have investigated whether tumors induce apoptosis in dendritic cells (DC) by co-culture of murine or human DC with different tumor cell lines for 4–48 h. Analysis of DC morphological features, JAM assay, TUNEL, caspase-3-like and transglutaminase activity, Annexin V binding, and DNA fragmentation assays revealed a time- and dose-dependent induction of apoptosis in DC by tumor-derived factors. This finding is both effector and target specific. The mechanism of tumor-induced DC apoptosis involved regulation of Bcl-2 and Bax expression. Double staining of both murine and human tumor tissues confirmed that tumor-associated DC undergo apoptotic death *in vivo*. DC isolated from tumor tissue showed significantly higher levels of apoptosis as determined by TUNEL assay when compared with DC isolated from spleen. These findings demonstrate that tumors induce apoptosis in DC and suggest a new mechanism of tumor escape from immune recognition. DC protection from apoptosis will lead to improvement of DC-based immunotherapies for cancer and other immune diseases. *J. Leukoc. Biol.* 66: 336–344; 1999.

Key Words: apoptosis · immunosuppression · Bcl-2

INTRODUCTION

Tumor-induced immunosuppression and the resulting progressive growth of neoplasms have been recognized for many years. However, the underlying mechanisms are still poorly understood. Tumor escape from immune rejection may be due, among other causes, to decreased or absent expression of MHC molecules [1], co-stimulatory molecules [2], or secretion of immunosuppressive factors by tumors. Several tumor or host-derived substances have been reported to inhibit immune responses. These factors include transforming growth factor β (TGF- β), interleukin (IL)-10, VEGF, and IL-4, as well as prostaglandin E_2 (PGE $_2$), hydrogen peroxide (H $_2$ O $_2$), nitric oxide (NO), soluble IL-2 receptors, complement inhibitors, proteases, gangliosides, hexosamines, α -fetoprotein, fibronectin, and phosphatidylserine [3–6].

O'Mahony et al. [7] reported that immunosuppressive factors derived from human esophageal squamous carcinoma induced

apoptosis in normal and transformed Jurkat T cells. Inhibition of the immune system by tumors can also be mediated by killing of immune cells within the tumor microenvironment. Tumor-induced apoptosis of T cells has been described in virtually all tumors that have been carefully examined, including melanoma, colorectal cancer, hepatocellular carcinoma, breast cancer, lung carcinoma, astrocytoma, and neuroblastoma [8]. In addition, the total number of natural killer (NK) cells is also decreased substantially in cancer patients [9], suggesting that tumor-induced apoptosis of NK cells might contribute to the inhibition of antitumor immune responses observed in cancer patients and tumor-bearing animals.

The initial definition of apoptosis was based on cellular morphology: condensation of chromatin and DNA, blebbing and shrinkage of cytoplasm, and eventual disintegration of dead cells within membrane-bound apoptotic bodies [for review see ref. 10]. More recently, other biochemical measures of apoptosis have been identified which include the internucleosomal cleavage of DNA into oligonucleosomal fragments of 180- to 200-bp multiples [11] and activation of intracellular cysteine proteases. Several families of enzymes have been shown to regulate apoptosis. Caspase activation is associated with the early stages of apoptosis [see review in ref. 12]. At least 10 different caspases have been identified in mammals. Caspases are categorized into three subgroups: ICE-like (caspase-1), CPP-32-like (caspase-3), and Ich1-like proteases [13]. Tissue transglutaminase (tTG), or type II transglutaminase, belongs to a family of enzymes that catalyze protein cross-linking reactions by creating ϵ -(γ -glutamyl)lysine bonds between the γ -carboxamide group of a glutamine residue in one polypeptide chain and the ϵ -amino group of a lysine residue in a second polypeptide chain. In mammals, tTG is among those specifically induced during the program of apoptosis [14]. In apoptotic cells, the activation of tTG by p53, glucocorticoids, retinoic acid, nur77, or TGF- β results in the assembly of a highly cross-linked protein structure that presumably prevents the

Abbreviations: DC, dendritic cell; TGF- β , transforming growth factor β ; IL-10, interleukin-10; PGE $_2$, prostaglandin E_2 ; NO, nitric oxide; NK, natural killer; tTG, tissue transglutaminase; rh, recombinant human; GM-CSF, granulocyte-macrophage colony-stimulating factor; FCS, fetal calf serum; 2-ME, 2-mercaptoethanol; UV, ultraviolet; ELISA, enzyme-linked immunosorbent assay; PBS, phosphate-buffered saline; SDS, sodium dodecyl sulfate; PMSF, phenylmethylsulfonyl fluoride; HRP, horseradish peroxidase; DAB, diaminobenzidine; LC, Langerhans cell; TIL, tumor-infiltrating leukocytes.

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release of harmful intracellular components from the dying cell, thereby avoiding an inflammatory reaction in neighboring tissues. In addition, it has been recently suggested that tTG may have another specific function in the regulation of apoptosis [14].

The aim of this study was to investigate whether tumors induce apoptosis of DC. Because apoptotic death can occur without caspase, transglutaminase activation, or DNA fragmentation [12, 15–17], we evaluated tumor-induced apoptosis of DC by assessing several measures of apoptosis. We demonstrate here that both murine and human tumors induce apoptosis in DC both *in vivo* and *in vitro*.

MATERIALS AND METHODS

Animals

Male C57BL/6 mice, 6–8 weeks old, were obtained from Taconic (Germantown, NY) and housed in a pathogen-free facility under controlled temperature, humidity, and a 12-h light:dark cycle with food and water available *ad libitum*. All animals were acclimatized for at least 2 weeks before the experiments. All experimental protocols were reviewed and approved by the University of Pittsburgh Institutional Animal Care and Use Committee.

Dendritic cell culture

Murine DC were generated as described previously [18]. Human DC were obtained from adherent peripheral blood mononuclear cells cultured with recombinant human granulocyte-macrophage colony-stimulating factor (rhGM-CSF; 1000 U/mL) and rhIL-4 (1000 U/mL) for 7 days in complete RPMI-1640 medium supplemented with 10% heat-inactivated Dextran-coated charcoal-depleted fetal calf serum (FCS). Human DC were also generated from CD34⁺ precursor cells isolated from the peripheral blood using CEPRATE LC affinity columns (CellPro, Inc., Bothell, WA). CD34⁺ precursors were cultured for 14 days in AIM V medium (BioWhittaker, Walkersville, MD) supplemented with 2.5% heat-inactivated human AB serum and the following cytokines: rhGM-CSF, 1000 U/mL; rhIL-4, 1000 U/mL; recombinant human tumor necrosis factor α (rhTNF- α), 2.5 ng/mL; and rhFLT3-ligand, 100 ng/mL.

Tumor cell lines

The following murine tumor cell lines were used: B16 and CL8-1 melanomas, C3 sarcoma, MC38 colon carcinoma, and TS/A mammary adenocarcinoma. MEL-526 melanoma, BE neuroblastoma, and Hep-1 hepatoblastoma cell lines were used to assess the apoptosis of human DC. Cell lines were expanded in an RPMI-1640-based complete medium supplemented with 5% heat-inactivated FCS, 0.1 mM non-essential amino acids, 1 mM sodium pyruvate, 2 mM glutamine, 50 μ M 2-mercaptoethanol (2-ME), 100 U/mL penicillin, and 100 mg/mL streptomycin. All cell cultures were mycoplasma-free.

Co-culture design

DC (1 to 2 $\times 10^6$ cells/mL) and individual tumor cell lines (1 $\times 10^6$ cells/mL) were co-cultured using 0.4- μ m pore size inserts (Falcon) in six-well plates (Falcon). The inserts containing tumor were removed after 4–48 h. In some experiments, inserts were removed after 24 h of culture, and the remaining DC were cultured for an additional 24 h.

Measurement of DNA fragmentation

Agarose-gel electrophoresis

DC were co-cultured with tumor cells or splenocytes as a control, harvested, and lysed in the 50 mM Tris-HCl buffer (pH 8, 20°C) containing 0.5% sarcosyl and 10 mM EDTA. DC were digested with proteinase K (0.5 mg/mL) in a total volume of 50 μ L at 50°C for 1 h followed by the incubation with 10 μ L of 10 mg/mL DNase-free RNase. Ten microliters of each sample in a loading buffer were loaded in the wells of 2% agarose gel and allowed to solidify. The gel was

run in TAE buffer containing ethidium bromide and visualized under ultraviolet (UV) light.

Cell death enzyme-linked immunosorbent assay (ELISA)

Degrees of DNA fragmentation were also measured using Cell Death Detection ELISA (Boehringer-Mannheim, Indianapolis, IN).

JAM assay

[³H]DNA release, or JAM assay, was performed as described [19] with minor modifications. Briefly, DC were pulsed with 3 μ Ci/mL [³H]thymidine (6.7 Ci/mmol, NEN, Boston, MA) in complete medium supplemented with 1000 U/mL rmGM-CSF and 1000 U/mL rmIL-4 for 24 h, washed twice, and loaded onto Nycoprep cell separation solution (1:1 mixed 1.068 and 1.077 g/mL, Nycomed Pharma AS, Oslo, Norway). After centrifugation for 15 min at 400 g, cells from the interface exhibited a viability of >99%. [³H]thymidine-labeled DC (10⁵) were mixed with tumor cells (DC/tumor cell ratios varied from 1:0 to 1:40) in round-bottom 96-well plates in a total volume of 200 μ L. Four to twenty-four hours later, cells were harvested onto the GF/C glass fiber filter with a MACH III harvester (Tomtec, Hamden, CT). Intact DNA within nuclei was retained by the filter, whereas the fragmented DNA was washed away. Remaining radioactivity was determined using a 1450 MicroBeta Trilux Counter (WALLAC, Gaithersburg, MD) and expressed as cpm. The percentage of DNA fragmentation was calculated as follows: %DNA fragmentation = (cpm_{control} – cpm_{experimental})/cpm_{control} $\times$ 100.

Morphological analysis

One hundred microliters of DC suspension were loaded into a cytospin chamber and spun for 5 min at 500 rpm. Slides were air-dried at room temperature for 5 min and stained in a three-step procedure using the LeukoStat Stain Kit (Fisher, Pittsburgh, PA) or a Wright-Giemsa Stain Kit (Shandon Lipshaw, Pittsburgh, PA). One hundred cells in three separate fields were counted, noting the percentage of apoptotic, necrotic, and viable cells based on the following criteria: cell membrane convolutions, e.g., blebbing, nuclei shrinkage and chromatin condensation, cytoplasmic constriction with a reduction in cell volume, formation of apoptotic bodies (for apoptotic cells), or chromatin flocculation, nuclear swelling, and cell membrane lysis resulting in ghost cells (as a measure of necrotic cells).

Immunoblotting

For the preparation of cell lysates, DC were washed three times in phosphate-buffered saline (PBS) and lysed using 50 mM Tris-HCl detergent buffer (pH 8.0) containing 1% Triton X-100, 0.1% sodium dodecyl sulfate (SDS), 150 mM NaCl, 0.5 mM phenylmethylsulfonyl fluoride (PMSF) at 4°C for 30 min. The cell lysates were clarified by centrifugation. Protein concentration was determined by the Bradford method [20] using the Bio-Rad Protein Assay (Bio-Rad Laboratories, Hercules, CA). Lysates were resolved on 10% SDS-polyacrylamide gels. Proteins were electrophoretically transferred to PVDF membranes (Millipore, Bedford, MA) overnight at 4°C. To detect Bcl-2 and Bax, the membranes were first blocked in a 5% (w/v) solution of non-fat milk in TBST buffer (150 mM NaCl, 0.05% Tween-20, and 10 mM Tris-HCl, pH 7.6) for 1 h at room temperature. Membranes were then incubated with the primary antibody (Santa Cruz Biotechnology, Santa Cruz, CA), diluted 1:200 in blocking buffer, for 60 min at room temperature. The blots were washed three times in TBST buffer and incubated with anti-rabbit secondary Ab conjugated with horseradish peroxidase (HRP) for 1 h at room temperature. The blots were washed extensively, and the proteins were detected using the Western blot chemiluminescence reagent (New England Nuclear).

Immunohistochemistry

For the *in situ* immunohistochemical evaluation of DC apoptosis in tumor tissues, murine or human samples were snap-frozen in OCT Compound (Sakura Finetek, Torrance, CA). Six-micrometer slices were dried overnight, fixed in acetone for 10 min, and stained with the murine DC-specific Ab NLDC-145 (DEC 205; Serotec, Oxford, UK) as described earlier [18]. For the human DC staining, anti-CD83 Abs (Immunotech, Westbrook, ME) or anti-CD1a Abs (PharMingen, San Diego, CA) were used. Ab binding was localized by a biotinylated secondary Ab, avidin-conjugated HRP, and diaminobenzidine

(DAB) substrate using an ABC detection kit (Vector Lab., Burlingame, CA). Isotype-matched Abs were used as a negative control. For the detection of apoptotic DC, TUNEL assay was applied as described below.

Transglutaminase activity assay

This assay is based on the determination of enzymatic activity of tissue transglutaminase (tTG) by measurement of the incorporation of a [^3H]putrescine into a protein substrate dimethylcasein as described earlier [21]. The lysine side-chains of casein were blocked with methyl groups to prevent the cross-linking reaction between glutamyl and lysine residues. Cytosolic and particulate cell extracts were prepared as previously described [22].

Caspase-3 activity assay

Caspase-3-like activity in DC was determined using the ApoAlert CPP32 Assay Kit (Clontech, Palo Alto, CA). Cell lysates from 2 to 3×10^6 cells were prepared according to the manufacturer's protocol and the caspase-3-like activity was measured by the spectrophotometric detection of the chromophore *p*-nitroanilide (pNA) at 405 nm after caspase-3-catalyzed cleavage from the labeled substrate DEVD-pNA. The units of protease activity were quantified using a pNA calibration curve.

TUNEL assay

Detection of DNA fragments *in situ* was performed using a TdT-FragEL Kit (Oncogene Research Product, Cambridge, MA) using the terminal deoxynucleotidyl transferase (TdT)-mediated dNTP nick end labeling assay. In this assay, TdT binds to exposed 3'-OH ends of DNA fragments generated in response to apoptotic signals and catalyzes the addition of biotin-labeled and unlabeled deoxynucleotides. Biotinylated nucleotides were detected using a streptavidin-HRP conjugate. DAB was used as a substrate for labeled samples to generate an insoluble colored (brown) stain at the site of DNA fragmentation. Counterstaining with methyl green aided in the morphological evaluation of normal and apoptotic DC. Cytospin slides of DC preincubated with tumor cells were prepared as described above after fixation in 4% formaldehyde for 10 min at 20°C and 80% ethanol.

Reagents

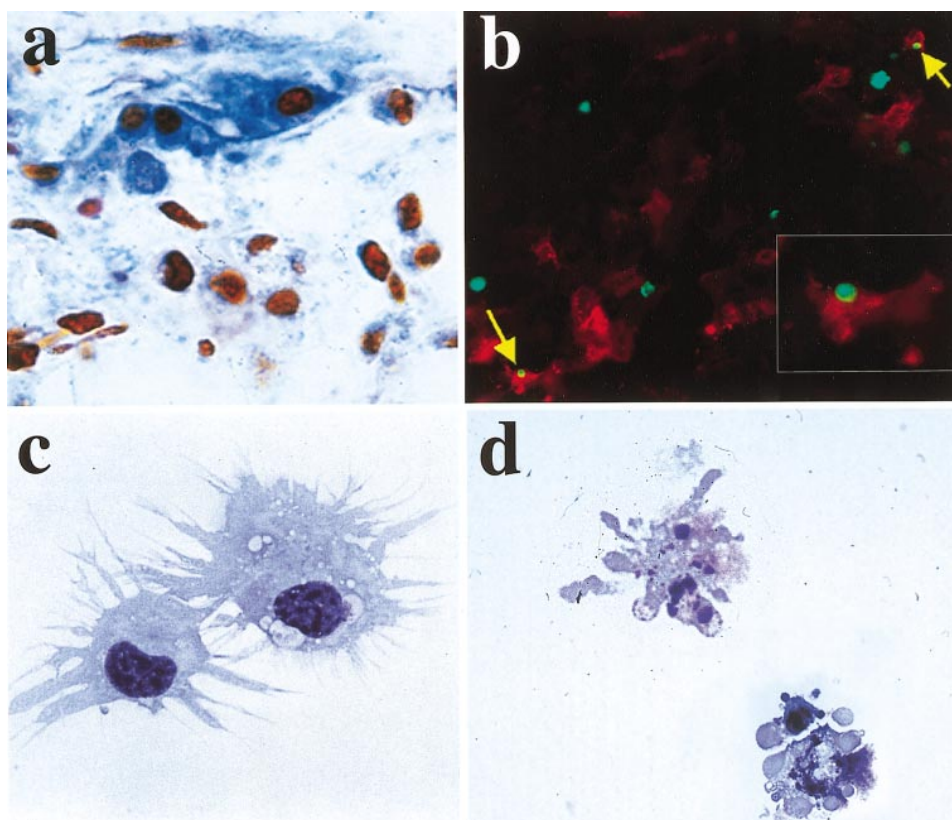
Sarcosyl, trichloroacetic acid (TCA), proteinase K, RNase, agarose, L-NAME, indomethacin, SDS, EDTA, dithiothreitol (DTT), dextran, Triton X-100, Tween 20, NaCl, PMSE, and charcoal (Norit A) were purchased from Sigma (St. Louis, MO). Tris, HEPES, 2-ME, non-essential amino acids, glutamine, sodium pyruvate, penicillin, streptomycin, gentamicin, and RPMI-1640 were obtained from GIBCO-BRL (Gaithersburg, MD). Antibodies used for flow cytometry were purchased from PharMingen (San Diego, CA). Formaldehyde was from Fisher (Pittsburgh, PA). rhGM-CSF and rhIL-4 were a gift from Schering-Plough Research Institute (Kenilworth, NJ), TNF- α was obtained from Knoll Pharmaceuticals (Whippany, NJ), and FLT3 ligand was a gift from Immunex Corp. (Seattle, WA).

RESULTS

Detection of apoptotic DC in human and murine tumors

To determine whether DC would undergo apoptosis within the tumor microenvironment *in vivo*, we first identified apoptotic DC in snap-frozen tumor sections with the use of immunohistochemical double staining. DC were visualized using NLDC-145 or anti-CD83 Abs for murine or human samples, respectively. Apoptosis was assessed by TUNEL assay. In order to decrease the number of apoptotic T cells, CL8-1 melanoma, and MC38 colon adenocarcinoma were grown in T cell-deficient nude mice and harvested when tumors reached sizes of 30–40 mm². Immunohistochemical analysis revealed the presence of apoptotic DC (data not shown). Also, CD83⁺ TUNEL⁺ double-positive apoptotic DC were detected within human tumors (Fig. 1, a and b). These results were confirmed using

Fig. 1. (a and b) Double staining of human melanoma (a) and ovarian cancer (b) by a combination of TUNEL assay (brown color in panel a or green in panel b) and staining of the cell surface antigen CD83 (blue color in panel a) or CD1a (red color in panel b) revealed the presence of apoptotic DC (arrows). Inset: original magnification $\times 600$ of apoptotic DC demonstrating green apoptotic nuclei surrounded by red CD1a staining in cytoplasm. (c and d) Morphological changes in murine bone marrow-derived DC after co-culture with B16 melanoma for 48 h (staining of cytospin specimens, original magnification $\times 1000$, d) compared to control DC incubated with splenocytes (c). Apoptosis was characterized by morphological features such as condensation of cytoplasm, cell shrinkage, membrane blebbing, chromatin condensation, and nuclear fragmentation.



anti-CD1aAbs. We next isolated CD11c⁺ DC from CL8-1 melanomas and spleens using MACS paramagnetic microbeads [23]. TUNEL staining of CD11c⁺ DC being freshly isolated from tumor tissue revealed significantly more apoptotic DC compared with splenic DC ($P < 0.05$). Thus, DC undergo apoptosis at the tumor site in both mice and humans.

Light microscopy reveals apoptosis in DC

Murine DC underwent apoptosis after pre-incubation across membranes with poorly immunogenic B16 melanoma cells for 24 h. Typical features of apoptosis included condensation of cytoplasm and shrinkage of the cell, membrane blebbing, chromatin condensation, and nuclear fragmentation. More than 90% of DC showed characteristics of apoptosis after co-culture with B16 cells for 48 h (Fig. 1, c and d). Co-culture with the highly immunogenic C3 sarcoma for 48 h resulted in only 20–25% of apoptotic DC. This finding suggests a tumor-specific killing. Freshly isolated splenocytes or phytohemagglutinin (PHA)-stimulated blasts served as controls and induced DC apoptosis rates below 10%. Thus, tumor cell-derived factors induced apoptotic death of cultured DC.

We next removed tumor cells after 24 h of co-culture and kept the DC alone for an additional 24 h. Results were identical to prior experiments. Thus, sufficient numbers of apoptotic signal(s) are released from tumor cells in the early stage of co-incubation. Furthermore, neither proliferation of tumor cells nor depletion of nutrients within the culture medium was responsible for the DC apoptosis. In conclusion, tumor-derived factors induced marked levels of apoptosis in human DC.

Live videomicroscopy allowed us to directly visualize alterations of DC morphology and the development of apoptotic bodies from a single DC following the contact with the tumor cell [8]. DC membranes were labeled using DiI/Cy3 (red) and DC nuclei were labeled with Hoechst 33342 (blue). One of the earliest tumor-induced changes of DC morphology was the disappearance of veils followed by nuclear and cytoplasmic condensation and blebbing. Contact and attachment to a tumor cell resulted in DC death within several hours. Thus, direct cell/cell contact resulted in earlier DC death compared to the effect of soluble tumor-released factors alone.

[³H]DNA fragmentation assay reveals DNA cleavage in DC

We first established 8 h of co-incubation as the optimal condition for the JAM assay, providing (1) the lowest background release of radioactivity and (2) the maximum tumor-induced release of [³H]DNA with the lowest level of secondary incorporation of [³H]DNA into proliferating tumor cells. As shown in **Figure 2**, co-culture of four different murine tumor cell lines with [³H]thymidine-labeled murine DC resulted in a dose-dependent decrease in viable cells with up to 90% diminution for the highest tumor/DC ratio. Co-culture with fibroblasts or freshly isolated splenocytes did not induce significant DC death. The percentage of DNA fragmentation in [³H]DC induced by splenocytes never exceeded 20% at the highest ratio, which was significantly lower than tumor-induced DC killing ($P < 0.001$). These findings suggest that tumor/DC

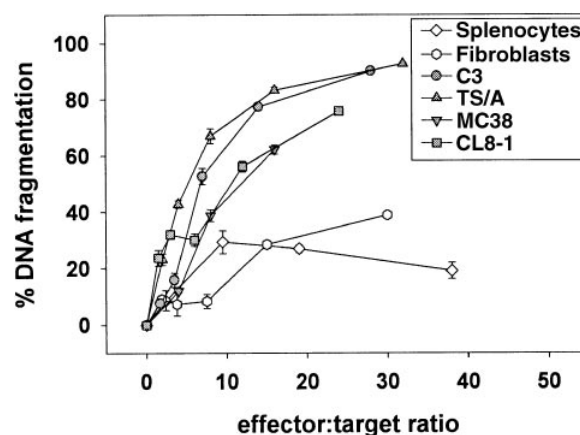


Fig. 2. Evaluation of DNA fragmentation in murine DC by a JAM assay. DC were labeled with [³H]thymidine at a concentration of 3 μ Ci/mL. After 24 h, excess [³H]thymidine was washed off, dead cells were removed by density centrifugation, and DC were co-incubated with tumors or control cells at E/T ratios up to 32:1 in 96-well round-bottom plates. Eight hours later, cells were harvested and the levels of intact DNA were detected. Co-incubation with five different tumors resulted in significant killing of DC compared to incubation with fibroblasts or splenocytes. Results are the mean of triplicate counts $\pm$ SEM of a representative experiment.

contact and/or release of soluble tumor-derived factors promote murine DC death *in vitro*.

The impact of tumor cells on the survival of human DC was also tested. DC were generated from either monocytes or isolated CD34⁺ precursors and labeled with [³H]thymidine for 48 h before incubation with tumor cells. MEL-526 melanoma cells induced a significant and dose-dependent reduction of viable DC numbers in both monocyte and CD34-derived DC. **Figure 3** shows a representative experiment using monocyte-derived DC. Human hepatoblastoma cells induced less DNA fragmentation in DC, although they induced a marked apoptotic death of Jurkat T cells (data not shown). This suggests a

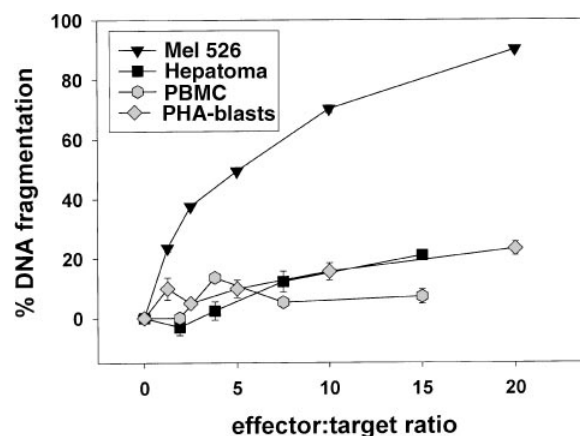


Fig. 3. Evaluation of DNA fragmentation in human monocyte-derived DC by a JAM assay. The experimental procedure is described in the legend for Figure 2. Results are expressed as the mean of triplicate counts $\pm$ SEM of a representative experiment. DC exhibited more than 70% DNA fragmentation after co-incubation with MEL 526 cells. In contrast, less than 10% DNA fragmentation was observed in DC co-incubated with PBMC. Monocyte-derived DC exhibited a lower uptake of [³H]thymidine than CD34-derived DC. Nevertheless, the percentages of DNA fragmentation in monocyte-derived and CD34-derived DC were comparable (see text).

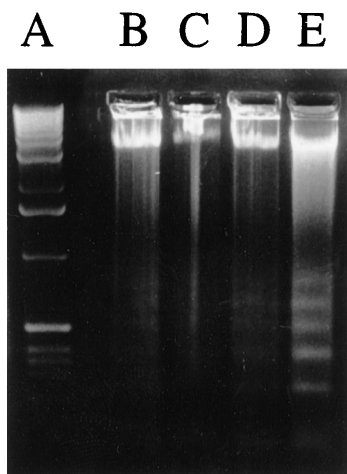


Fig. 4. Agarose gel electrophoresis of DNA extracted from murine DC after co-incubation with B16 melanoma, C3 sarcoma, or controls. Ten micrograms DNA per lane were run on a 2% agarose gel stained with ethidium bromide. DNA fragmentation was visualized by a UV (302 nm) transilluminator. Analysis of these data showed the characteristic laddering of DNA after co-incubation with B16 melanoma cells (lane E) but not C3 sarcoma (lane D), PHA-activated blasts (lane C), or medium (lane B). Lane A, DNA size markers.

target-specific mechanism of tumor-induced apoptosis of immune cells: different cell types exhibit different levels of sensitivity to tumor-induced death, however, DC subtypes such as monocyte-derived and CD34-derived DC, are equally sensitive.

DNA fragmentation in DC by electrophoresis, ELISA, and TUNEL assays

To further characterize tumor-induced DNA fragmentation and to distinguish between apoptotic and necrotic DNA degradation, we collected DC at different timepoints during co-culture with tumor cells and separated DNA fragments by electrophoresis on an agarose gel. DNA laddering was detected in the samples extracted from murine DC co-incubated with B16 melanoma cells for 48 h (**Fig. 4, lane E**). Figure 4, lane D, illustrates that co-incubation with C3 sarcoma cells did not result in DNA laddering. These data suggest an effector-specific mechanism of tumor-induced DC death.

We never observed DNA laddering patterns at early timepoints (<24 h) of DC/tumor cell co-cultures in inserts (data not shown). This may be due to the limited sensitivity of agarose-gel electrophoresis of isolated DNA to assess internucleosomal cleavage. To determine whether oligonucleosomal cleavage occurred in DC earlier than 24 h of pre-incubation with tumor cells, we evaluated the presence of mono- and oligonucleosomes in the cytoplasmic fraction of DC lysates by a quantitative sandwich enzyme immunoassay using mAbs directed against DNA and histones, respectively. DC/B16 co-cultures were initialized as described above and DC were harvested and analyzed 8 h later. We found that the OD (410 nm), which reflects release of mono- and oligonucleosomes into the cytoplasm, increased from 0.579 ± 0.003 U in control samples (DC cultured without tumor cells) to 0.696 ± 0.006 U in samples obtained from B16 tumor co-cultures ($P < 0.001$). Thus, this observation confirms the finding of tumor cell-induced apoptosis of DC *in vitro*.

Further evidence of tumor-induced DNA fragmentation in DC at an early time of co-incubation was obtained using TUNEL techniques. Murine DC were collected after 8 h of pre-incubation with membrane-separated B16 cells, washed, fixed, spun onto slides, and stained as described in Materials and Methods. The number of TUNEL-positive DC significantly increased if tumor cells were added to DC cultures. For instance, morphometric analysis revealed a threefold increase in apoptotic cell number in DC cultures pre-incubated with B16 cells compared with control cultures: 16 ± 3 vs. $6 \pm 2\%$, respectively ($P < 0.01$). Thus, detection of DNA fragmentation *in situ* through the end-labeling of double-strand DNA breaks using TdT also supports our finding of tumor-induced apoptosis of DC.

In summary, direct contact with a tumor can result in measurable DNA fragmentation in DC after only a couple of hours.

Caspase-3 is activated in apoptotic DC

Caspase-3 is widely distributed with high levels of expression in cells of lymphocytic origin and is an important mediator/ marker of apoptosis in the immune system [24]. We measured caspase-3-like activity in DC at different time-points after pre-incubation with a variety of tumor cells. **Figure 5** shows the results of a representative experiment, demonstrating caspase-3 activity in the DC cultures as determined by the cleavage of chromophore pNA from the labeled peptide substrate DEVD-pNA. Murine DC were pre-incubated with B16 melanoma, C3 sarcoma, MC38 adenocarcinoma cells, freshly isolated splenocytes, PHA-blasts (24-h stimulated with 2.5 μ g/mL PHA splenocytes), or medium for 24 h. Inserts, containing tumor or control cells, were removed and the aliquots from

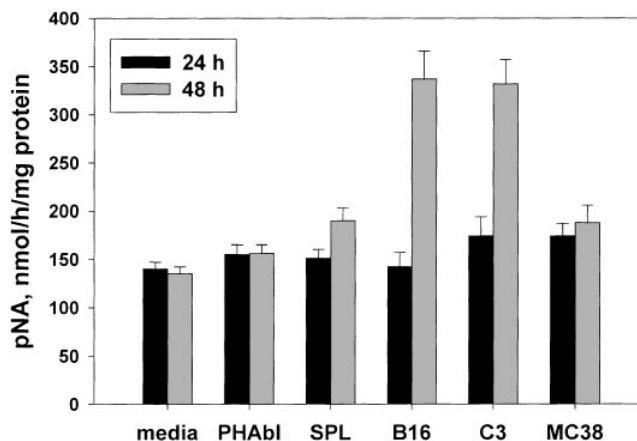


Fig. 5. Caspase-3-like activity in murine DC after co-incubation with different effector cells. Cultured murine DC were preincubated with B16 melanoma, C3 sarcoma, MC38 adenocarcinoma cells or freshly isolated splenocytes, PHA blasts (24-h stimulated with 2.5 μ g/mL PHA splenocytes), or medium for 24 h. Inserts, containing tumor or control cells, were then removed and aliquots of cultured DC were collected, washed, and frozen at -20°C . All DC cultured were incubated for an additional 24 h (to obtain 48-h time point) and then harvested, washed, and frozen. Results are the mean of triplicate measurements $\pm$ SEM for a representative experiment. B16 melanoma and C3 sarcoma caused a significant elevation of caspase-3 activity in comparison with control DC, whereas MC38 colon carcinoma, splenocytes, or PHA blasts resulted only in a slight increase in caspase-3 activity.

DC cultures were washed and frozen at -20°C . DC were incubated for an additional 24 h and then washed and frozen. Only B16 and C3 cells induced a significant elevation of caspase-3-like activity (Fig. 5), whereas other cells, including MC38 carcinoma, induced only a slight increase.

In addition, we evaluated the caspase-3-like activity in DC cultured with B16 or C3 cells for 48 h without interruption. We confirmed that both tumor cell lines activated caspase-3 in DC. The dynamic of this activation was slightly different from that shown in Figure 5. In fact, we observed lower levels of the C3-induced caspase-3 activation at 48 h in cultures where tumor cells were not removed, compared with activation levels in cultures where C3 cells were displaced after 24 h of co-culture: 207.2 ± 6.9 vs. 332.5 ± 15.2 nmol/h/mg, respectively. No differences were observed for the B16-induced caspase-3 activation in DC: 342.8 ± 11.4 vs. 337.1 ± 15.7 nmol/h/mg. These results should reflect differences in levels of tumor-derived factors. Moreover, decrease in C3-induced caspase-3 activation after removal of tumor cells suggests that the effect of C3 but not B16 cells can be reversed.

To evaluate the involvement of caspase-3 in the apoptosis of human DC, 0.75×10^6 MEL526 cells or peripheral blood mononuclear cells (PBMC) were added in inserts to CD34-derived DC cultures for 48 h. Medium was used as an additional control. We observed an increase in caspase-3-like activity in DC preincubated with tumor cells: 838.6 ± 56.8 nmol/h/mg compared with 576.7 ± 43.2 and 564.4 ± 35.9 nmol/h/mg in DC cultured with PBMC and medium, respectively. Thus, tumor-induced apoptosis of both murine and human DC involved induction of caspase-3 proteolytic activity in these cells.

Transglutaminase activation during DC apoptosis

Activation of tTG correlates with the induction of apoptosis in various cell populations [14]. We have determined TG activity in DC samples obtained as described above. Addition of B16 or C3 tumor cells to the DC cultures resulted in a statistically significant, time-dependent increase in tTG activity in DC. Unexpectedly, B16 cells did not induce a marked activation of tTG activity in DC, suggesting that the tTG activation in DC might be tumor-specific. To determine the reversibility of the tumor effect on tTG activation in DC, C3 and B16 cells were removed 24 h after the initiation of co-cultures and DC samples were harvested and analyzed 24 h later. We found that the C3-stimulated tTG activity in DC was significantly lower in these samples compared with the DC tTG activity measured in corresponding controls ($P < 0.05$). These results suggest that both the level and dynamics of activation of tTG activity in DC were tumor-specific.

Involvement of Bcl-2 protein family in tumor-induced apoptosis in DC

The balance between Bcl-2 inducers and repressors regulates apoptotic cell death in many cell types [25]. We found that pre-incubation of murine bone marrow-derived DC with B16 melanoma cells for 24 h resulted in significant down-regulation of Bcl-2 expression in DC. In contrast, the expression of Bax protein in the same DC was up-regulated (Fig. 6). These results

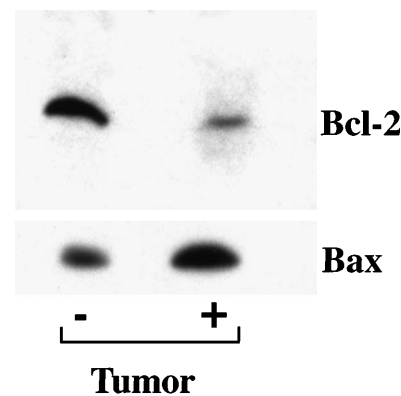


Fig. 6. Western blot analysis of Bcl-2 and Bax expression in DC pre-incubated with B16 melanoma cells. After incubation with tumor cells or medium alone for 24 h, DC were harvested, lysed, and Western blot analysis was performed as described in Materials and Methods. Each lane was loaded with 40 μg total protein. These data demonstrated marked inhibition of Bcl-2 expression and activation of Bax expression in DC induced by tumor-derived factors.

demonstrate involvement of the Bcl-2 family of proteins in the regulation of tumor-induced apoptosis of DC and suggest that Bcl-2 is important for DC survival. Bcl-2 might represent a target for strategies designed to protect DC from tumor-induced apoptosis.

DISCUSSION

We have demonstrated that both murine and human DC undergo apoptosis *in vitro* and *in vivo* after contact with tumors. Tumor growth is associated with inhibition of the immune response in both humans and animals [26]. One mechanism of tumor-associated immunosuppression is inhibition of immune effector cells at the tumor site, including T cells and macrophages. There is increasing evidence that DC function can also be modified by tumor-derived factors.

DC initiate and regulate anti-tumor immune responses [27, 28]. Impairment of DC number or activity might result in deficient expansion or activation of specific T lymphocytes. In fact, alterations in the density and distribution of Langerhans cells (LC) within the epidermis and in the peritumoral infiltrate of malignant melanoma might determine the degree of T cell activation [29]. Recent studies reported on tumor-induced inhibition of DC function or differentiation. Human CD83⁺ DC, obtained from progressing, chemotherapy-resistant melanomas, revealed a marked down-regulation of CD86 and induced anergy in syngeneic CD4⁺ T cells [30]. This anergy could be overcome by supplementation with IL-12. Gabrilovich et al. demonstrated that DC from breast cancer patients failed to stimulate proliferation of allogeneic T lymphocytes [31]. Similar data were obtained using human basal cell carcinoma-associated DC, which are deficient in CD80 and CD86 expression as well as in their ability to stimulate T lymphocyte proliferation [32]. The inability of tumor-associated DC to effectively stimulate T cells may explain why tumor-infiltrating lymphocytes (TIL) fail to eliminate tumor cells. Tumor-derived factors inhibiting DC function still await isolation. Substances known to affect T cells, macrophages, and NK cells could alter

DC activity as well. VEGF, produced by a variety of tumor cells, has been shown to inhibit the differentiation of DC from CD34⁺ progenitor cells [5]. PGE₂ is also able to prevent the generation of CD1⁺ DC from peripheral blood precursors [33].

We have described a new mechanism that is likely to contribute to tumor-mediated immunosuppression. Apoptotic death of DC has been previously described for a number of inflammatory states. Infection of human CD34-derived DC by measles virus induced 25% of DC to undergo apoptosis after 4 days of infection, as detected using Annexin V and propidium iodide staining [34]. Another group, using TUNEL technique, has demonstrated that in measles virus-infected monocyte-derived human DC cultures ~45% of cells undergo apoptosis by the conclusion of culture [35]. In contrast, UV-inactivated measles virus did not induce apoptosis of DC, suggesting that this death was largely due to viral replication. It is likely that members of the TNF family are up-regulated on DC after viral infection. This might induce paracrine killing of T cells and autocrine killing of DC cultures [35]. The virus, both infectious and UV-inactivated, was able to inhibit IL-12 production by CD40-activated DC [35]. This phenomenon contributes to the Th2 polarization observed in virus-infected patients. HIV infection might also induce functional polarization of Th2 responses.

Tumor progression has also been associated with the modifications of the Th1/Th2 cytokine balance. As tumors progress, a Th1 response may shift to a Th2-dominant response, which could result in down-regulation of CD8⁺ T cell function and consequent decreased ability to generate anti-tumor effector cells [36]. In fact, both gradual diminution of Th1 cells as well as IL-12 production has been described in tumor-bearing hosts [37–40]. On the other hand, DC are an important source of IL-12 and this cytokine is likely to be involved in the anti-tumor activity of DC [41]. Taken together with our observation of tumor-induced DC apoptosis, these findings suggest an additional IL-12-related mechanism of immunosuppression. Dysfunction and death of DC in the tumor microenvironment would result in a marked deficiency of IL-12 synthesis and could explain the shift from Th1 to Th2. Furthermore, utilizing UV radiation as an apoptotic signal, Kitajima et al. [42] suggested that DC undergoing apoptosis deliver unusual activation signals to T cells during antigen presentation, signals that lead to cellular unresponsiveness rather than to effective immunity.

Mechanisms and factors responsible for tumor-induced DC death have not yet been identified. Recent reports demonstrated FasL expression by various human and murine tumors, suggesting that Fas/FasL-mediated apoptosis of TIL might contribute to maintaining an immune privilege at the tumor site [43]. We have recently demonstrated that cultured murine DC express Fas mRNA and protein [8]. The majority of murine tumors used in this study expressed significant levels of intracellular FasL (data not shown). However, we did not observe functional involvement of the Fas/FasL interaction in melanoma-induced apoptosis of murine DC (data not shown). von Stebut et al. demonstrated LC apoptosis in human cultures that engaged Fas signaling pathways [44]. Thus, DC express both Fas and FasL, however, the involvement of this pathway in

apoptotic death could be effector-specific. We did not observe significant levels of DC apoptosis when the same DC were used as effector cells. It is likely that the Fas-mediated pathway is inhibited in murine DC, although FasL expression may still be functional. Lu et al. [45] reported that murine myeloid DC express FasL, which in association with B7 expression might regulate T cell survival. Also, a subpopulation of murine lymphoid CD8⁺ splenic-derived DC has been shown to induce apoptotic death of activated CD4⁺ T lymphocytes [46]. By using mixtures of T cells and DC from Fas-knockout *lpr/lpr* mice and FasL-knockout *gld/gld* mice, Suess and Shortman [47] were able to demonstrate that this death was due to the interaction of Fas on activated T lymphocytes with FasL on lymphoid DC. Thus, Fas/FasL interaction in tumor-induced DC apoptosis requires further clarification.

NO could also be considered as a tumor- or macrophage-derived factor being able to induce DC apoptosis. In fact, exposure of DC to the NO donor, SNAP, promoted apoptosis [48]. Another candidate is IL-10, which is produced by a variety of tumor cells [4]. Ludewig et al. demonstrated that IL-10 induces apoptosis of DC *in vitro* and reverses the effect of TNF- α and TRAP (CD40L), which inhibits spontaneous DC apoptosis in cultures [49]. We have recently found that CD40 ligation protected DC from tumor-induced apoptosis [50]. TNF- α also mediated enhanced resistance of murine DC to B16 melanoma-induced apoptosis *in vitro* [Esche and Shurin, unpublished observations]. Thus, mature DC do have a survival advantage within the tumor microenvironment. This finding suggests the use of mature DC for clinical trials.

Identification of mechanisms regulating DC apoptosis is necessary for further improvement of DC-based cancer vaccination [28]. Furthermore, infection-derived factors might also be of clinical interest. As mentioned above, infection of human DC by measles virus induced apoptosis [34]. Infection of murine DC by *Listeria monocytogenes* also promoted apoptotic death, suggesting a possible role of DC apoptosis in the pathogenesis of listerial infection [51]. However, DC also recognize apoptotic cells. Albert et al. recently demonstrated that apoptosis, but not necrosis, is required for the generation and packaging of immunogenic material for delivery to DC [52].

This study provides evidence for activation of caspases (CPP32) in DC during tumor-induced apoptosis. However, a single assay such as measurement of caspase-3 activity does not necessarily reflect the actual level of apoptosis directly. It is still an issue of discussion how well caspase activation correlates with apoptotic death and we therefore performed different assays in this study, although many data suggest a critical role for caspase-3 in programmed cell death [53, 54]. Tumor-induced apoptosis in DC was accompanied by down-regulation of Bcl-2 and up-regulation of Bax. This result suggests potential target molecules for attempts to increase DC survival within the tumor microenvironment. We have found tumor-induced DNA fragmentation in DC through the use of four different approaches. As expected, the TUNEL technique turned out to be more sensitive than agarose electrophoresis in detecting small amounts of DNA fragmentation. Also, direct contact between effectors and targets might result in earlier fragmentation events than co-incubation designs using inserts.

Our data demonstrate that mechanisms of tumor-induced apoptosis are both effector and target specific. Different machineries of apoptosis-inducing substances might be activated by different tumors. Thus, clinical trials for different tumors will require different DC protection approaches.

Intratumoral accumulation of DC is associated with prolonged survival and a reduced incidence of metastatic disease in patients with melanoma, oral, head, and neck tumors, nasopharyngeal cancer, lung, bladder, esophageal, and gastric carcinoma [55]. This phenomenon can be explained by our finding of tumor-induced DC death. More aggressive tumors induce higher levels of DC apoptosis at the tumor site and, in turn, stronger inhibition of antigen recognition, processing, and presentation by DC, which are necessary for the initiation and maintenance of an effective anti-tumor immune response [28]. In fact, Stene et al. reported that melanoma-associated Langerhans cells declined in number as melanoma progresses [56]. Determination of the frequency of epidermal LC in the epidermis overlying a primary melanoma revealed a substantial reduction of LC [57]. Identification of mechanisms regulating tumor-induced apoptosis of DC will facilitate approaches aimed to prolong DC survival.

The growth factor FLT3 ligand regulates hematopoiesis and anti-tumor immune responses [58]. Systemic administration induces tremendous stimulation of DC generation in mice [18, 59, 60] and inhibition of tumor growth in various murine tumor models, including B16 and CL8-1 melanomas, EL-4 lymphoma, C3 sarcoma, and MC38 colon carcinoma [61–63]. Based on the observation of tumor-induced apoptosis of DC, we hypothesize that the elimination of tumor-associated DC may permit tumor growth. On the other hand, enhanced numbers of DC within a tumor may mediate inhibition of tumor growth or tumor regression. Thus, the accumulation of intratumoral DC represents a new tool for immunotherapy of cancer. Prevention of apoptosis in intratumoral DC will become an additional strategy to improve the effectiveness of immunotherapy.

In summary, we demonstrated that tumor-derived stimuli induce apoptotic death of both murine and human DC. This finding is both effector and target specific. Tumor-induced death of DC is accompanied by the activation of caspase-3 and transglutaminase, increased expression of Bax, decreased expression of Bcl-2, and final formation of apoptotic bodies. This phenomenon represents a new mechanism of tumor-induced immunosuppression and escape from immune recognition. Effective protection of tumor-induced DC apoptosis and increased DC survival within the tumor microenvironment may markedly improve the efficacy of DC-based immunotherapies of cancer.

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